# Cherif BALI

## EMBEDDED SYSTEMS ENGINEER

City, Country • xxxx@email.com • +33 xx xx xx xx • https://www.linkedin.com/in/xxxx

**32 years old**
**6+ years of experience**
Sectors: Aerospace, Space, R&D


Embedded Systems Engineer specialized in low-level software development, integration, validation and test environments for critical systems. Experience in aerospace and industrial environments, with strong expertise in C/C++, debugging, Time analysis, RTOS and continuous integration.

## SKILLS

| | |
|---|---|
| C | ●●●●○ |
| Embedded Systems | ●●●●○ |
| Electronics | ●●●●○ |
| Teamwork | ●●●●○ |
| C++ | ●●●○○ |
| Python | ●●●●○ |

## KEY AREAS

**METHODOLOGIES**
Agile Scrum, V-Model, DO-178

**LANGUAGES**
C, C++, Python, Java, Shell/Bash

**COMMUNICATION PROTOCOLS**
AFDX, A429, A615, I2C, SPI, UART, CAN

**RTOS / RTK**
FreeRTOS, ASTERIOS

**TOOLS & DEBUG**
GDB, Trace32, RobotFramework, Jenkins, Gradle, IAR Workbench, VS Code, Eclipse, Git, SVN

**MODELING & SIMULATION**
MATLAB, Simulink

**AI / ML**
Machine Learning, Neural Networks, TensorFlow, Scikit-learn

**SYSTEMS**
Linux, Windows

## EDUCATION

**2017**
**Master's Degree in Electronics**
National Polytechnic School, Algeria

**2014**
**Engineering Studies**
Preparatory Classes for Grandes Écoles

## LANGUAGES

| | |
|---|---|
| French | Fluent |
| English | Professional |
| Arabic | Native |

## PROFESSIONAL EXPERIENCE

### Software Integration Engineer
11/2023 – Present
**AIRBUS**

Physical and simulated test environments for flight computers: tool adaptation, anomaly resolution (Jira), and technical support for avionics validation.

- **BSWTool project:** Porting of tests to real hardware targets, MFC test definition, debugging, documentation/PRF authoring and requirements traceability.
- **Test environment evolution:** Development of tools, including AFDXTool for handling avionics AFDX/Ethernet frames related to IFCC flight controls.
- **Support & Quality:** User support and optimization of test campaigns.

**Environment:** C, C++, DO-178, Java, RobotFramework, Git, Gradle, Jenkins, GDB, Trace32, CI/CD, PowerPC MCU, Jira, Airbus Process

### Embedded Research Engineer
02/2023 – 07/2023
**IRT SAINT EXUPÉRY**

Methods & Tools for Complex Systems department: software testing and timing analysis of a DAL B space application.

- Timing analysis and interference identification on a 6-core Infineon AURIX TriCore target.
- Code instrumentation, execution-time measurement and WCET estimation for DO-178 certification.

**Environment:** C, AURIX TriCore TC399 MCU, ASTERIOS RTK, AURIX IDE, Trace32 Profiling/Debugging, Shell Scripting, SCADE, DO-178

### Embedded Systems Engineer - R&D
02/2020 – 01/2023
**SAMSUNG Home Appliances (SAMHA)**

Design and development of embedded software for home appliances within a 15-person HW/SW R&D team.

- Development of bare-metal and RTOS embedded software.
- Implementation of low-level HAL/BSP layers for target boards.
- Analysis, debugging, unit and functional testing, and continuous integration.
- Electronic board testing/validation and cross-functional collaboration with Mechanical, Laboratory, Quality and Certification teams.
- Benchmarking, reverse engineering and measurement.

**Environment:** C, C++, Python, V-Model, Renesas MCU, STM32 MCU, RTOS, IAR IDE, SVN, Git, I2C, SPI, UART, HAL/BSP, PCB Design, SolidWorks